

Health Data Mapping, Insight Discovery and AI Readiness

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Data Source Catalogue

1. Apple Health

Apple Health categorizes its data primarily into `HKQuantityType` (numerical values) and `HKCategoryType` (small set of possible values, like sleep stages).

Cardiovascular and Vital Signs:

- **Heart Rate:** `heartRate`, `restingHeartRate`, `walkingHeartRateAverage`.
- **Heart Rate Variability (HRV):** Measured as the standard deviation of heartbeat intervals (`heartRateVariabilitySDNN`).
- **Blood Pressure:** Captures both `bloodPressureSystolic` and `bloodPressureDiastolic`.
- **Respiration & Oxygen:** `respiratoryRate`, `oxygenSaturation`, and `forcedVitalCapacity`.
- **Electrocardiogram:** `ElectrocardiogramType` for ECG data.
- **Body Temperature:** `bodyTemperature` and `basalBodyTemperature`.

Activity and Mobility Metrics:

- **Movement:** `stepCount`, `distanceWalkingRunning`, `distanceCycling`, and `distanceWheelchair`.
- **Energy:** `basalEnergyBurned` (resting) and `activeEnergyBurned`.
- **Advanced Gait Metrics:** `walkingSpeed`, `walkingStepLength`, `walkingAsymmetryPercentage`, and `appleWalkingSteadiness`.

Clinical and Wellness Data:

- **Body Measurements:** `bodyMassIndex` (BMI), `bodyFatPercentage`, `leanBodyMass`, and `waistCircumference`.
- **Sleep:** `sleepAnalysis` (category sample for sleep information).
- **Blood & Symptoms:** `bloodGlucose`, `bloodAlcoholContent`, and `electrodermalActivity` (sweat gland activity).

Summary Table:

Variable Name	Category
Heart Rate	Cardiovascular & Vitals
Resting Heart Rate	
Walking Heart Rate Average	
Heart Rate Variability	
Blood Pressure	
Electrocardiograms (ECG)	

Variable Name	Category
Respiratory Rate	Respiration & Oxygen
Blood Oxygen	
Body Temperature	Vitals / Cycle Tracking
Basal Body Temperature	
Steps	Activity & Mobility
Walking + Running Distance	
Resting Energy	
Active Energy	
Walking Speed	Advanced Gait Metrics
Walking Asymmetry	
Sleep	Clinical & Wellness
Blood Glucose	
Body Fat Percentage	Body Measurements
Lean Body Mass	

2. Samsung Health

Samsung Health explicitly tracks user-set goals as distinct, aggregatable data types.

Physiological and Sleep Metrics:

- **Cardiovascular:** Heart rate (supports aggregation and read changes), Blood pressure, and Irregular heart rhythm notifications.
- **Blood & Oxygen:** Blood glucose and Blood oxygen.
- **Sleep Data:** Tracks overall Sleep, Sleep apnea, and allows setting a “Sleep goal”.
- **Temperature & Composition:** Skin temperature, Body temperature, and Body composition.

Activity & Lifestyle:

- **Activity Tracking:** Exercise, Exercise location, and Floors climbed.
- **Energy & Nutrition:** Energy score, Active calories burned goal, Active time goal, Nutrition logs, and Nutrition goals.

Summary Table:

Variable Name	Category
Heart rate	Cardiovascular
Blood pressure	
Irregular heart rhythm notification	
Blood glucose	Blood & Oxygen
Blood oxygen	
Sleep	Sleep Data
Sleep apnea	
Sleep goal	
Skin temperature	Temperature
Body temperature	
Body composition	Composition

Variable Name	Category
Exercise	Activity Tracking
Floors climbed	
Energy score	Energy & Nutrition
Nutrition	

3. Google Health

Google specifically standardizes exercise distances (e.g. `elevationGainMillimeters`) in millimeters to maintain high precision across all developers, regardless of imperial or metric systems.

Cardiovascular & Vitals:

- **Heart Metrics:** Heart Rate, Resting Heart Rate, Daily Heart Rate Variability, Heart Rate Zones, and Electrocardiogram (ECG).
- **Respiratory & Oxygen:** Oxygen Saturation, Daily Respiratory Rate, and Core Body Temperature.

Activity & Body Metrics:

- **Energy & Workouts:** Active Energy Burned, Active Minutes, VO2 Max, Run VO2 Max, Steps, Distance, and Swim Lengths Data.
- **Body & Nutrition:** Weight, Height, Body Fat, Hydration Log, Food logs, and Nutrition Log.
- **Sleep:** Sleep sessions and Respiratory Rate Sleep Summary.

Summary Table:

Variable Name	Category
Heart Rate	Heart & Cardiovascular
Daily Resting Heart Rate	
Heart Rate Variability	
Electrocardiogram (ECG)	Vitals
Oxygen Saturation	
Daily Respiratory Rate	
Core Body Temperature	Activity & Workouts
Active Energy Burned	
VO2 Max	
Steps	Body Measurements
Distance	
Weight	
Height	Nutrition
Body Fat	
Nutrition Log	Sleep
Sleep	

4. Garmin

Garmin uniquely separates its data streams into All-Day Metrics (via the Health API/Companion SDK) and Real-Time Streaming Metrics (via the Standard/Companion SDKs).

All-Day Metrics (Historic/Daily Aggregations):

- **Energy & Stress:** Body Battery, Stress Levels, Calories, and Intensity Minutes.
- **Vitals:** Heart Rate, Pulse Ox, Respiration Rate, and Index Scale Biometrics.
- **Activity & Sleep:** Steps, Distance, Floors Climbed, Fitness Activity Details, Sleep, Advanced Sleep Data, and Health Snapshot.

Real-Time Streaming & Customizable Logging:

- **High-Resolution Physiological Data:** Enhanced Beat-To-Beat Intervals (crucial for accurate real-time HRV calculations), Heart Rate, Pulse Ox, and Respiration Rate.
- **Raw Kinematics:** Accelerometer (including sums of axis crossings), Gyroscope, and Actigraphy.
- **Device Status:** Device Battery Percentage and Unique Device ID.

Summary Table:

Variable Name	Category
Body Battery	Energy & Stress
Stress	
Intensity Minutes	
Calories	
Heart Rate	Vitals
Pulse Ox	
Respiration	
Body Composition	
Steps	Activity & Sleep
Floors Climbed	
Sleep	
Health Snapshot	

Data Dictionary

Below is a Data Dictionary mapping the different brand-specific inputs into standardized variables for each category.

1. Cardiovascular & Vitals

Ikarians Standard Variable	Data Type	Unit	Source Brand Mappings	Description
heart_rate	Numeric	bpm	Apple (Heart Rate), Samsung (Heart rate), Google (Heart Rate), Garmin (Heart Rate)	Continuous heart rate measurement.
resting_heart_rate	Numeric	bpm	Apple (Resting Heart Rate), Google (Daily Resting Heart Rate)	Baseline heart rate measured during rest.
heart_rate_variability	Numeric	ms	Apple (Heart Rate Variability), Google (Heart Rate Variability)	Variation in time intervals between heartbeats.
blood_pressure	Numeric	mmHg	Apple (Blood Pressure), Samsung (Blood pressure)	Captures systolic and diastolic measurements.
ecg_signal	Categorical	N/A	Apple (Electrocardiograms), Google (Electrocardiogram)	Electrocardiogram tracking and irregular rhythm notifications.

2. Respiration & Oxygen

Ikarians Standard Variable	Data Type	Unit	Source Brand Mappings	Description
respiratory_rate	Numeric	brpm	Apple (Respiratory Rate), Google (Daily Respiratory Rate), Garmin (Respiration)	Breaths per minute.
blood_oxygen	Numeric	%	Apple (Blood Oxygen), Samsung (Blood oxygen), Google (Oxygen Saturation), Garmin (Pulse Ox)	Peripheral oxygen saturation levels (SpO2).

3. Temperature

Ikarians Standard Variable	Data Type	Unit	Source Brand Mappings	Description
core_body_temperature	Numeric	°C/°F	Apple (Body Temperature), Samsung (Body temperature), Google (Core Body Temperature)	Estimated core physiological temperature.

4. Activity & Mobility

Ikarians Standard Variable	Data Type	Unit	Source Brand Mappings	Description
daily_steps	Numeric	Count	Apple (Steps), Google (Steps), Garmin (Steps)	Total step count over a specified interval.
active_energy_burned	Numeric	kcal	Apple (Active Energy), Google (Active Energy Burned), Garmin (Calories)	Caloric expenditure during movement.
distance_traveled	Numeric	meters	Apple (Walking + Running Distance), Google (Distance)	Standardized to millimeters by Google and tracked generally by Apple.
floors_climbed	Numeric	Count	Samsung (Floors climbed), Garmin (Floors Climbed)	Vertical movement tracking.
advanced_gait_metrics	Numeric	m/s, %	Apple (Walking Speed, Walking Asymmetry)	Speed, step length, and movement symmetry.

5. Sleep & Recovery

Ikarians Standard Variable	Data Type	Unit	Source Brand Mappings	Description
sleep_tracking	Categorical / Numeric	Minutes	Apple (Sleep), Samsung (Sleep), Google (Sleep), Garmin (Sleep)	Sleep duration and stage analysis.
sleep_apnea_flag	Boolean	True/False	Samsung (Sleep apnea)	Specific detection of sleep apnea events.
device_energy_score	Numeric	0-100	Samsung (Energy score), Garmin (Body Battery)	Device-specific proprietary recovery/energy algorithms.

Ikarians Standard Variable	Data Type	Unit	Source Brand Mappings	Description
device_stress_score	Numeric	0-100	Garmin (Stress)	Derived stress calculations based on HRV and activity.

6. Body Composition & Clinical Markers

Ikarians Standard Variable	Data Type	Unit	Source Brand Mappings	Description
blood_glucose	Numeric	mg/dL	Apple (Blood Glucose), Samsung (Blood glucose)	Endogenous blood sugar tracking.
body_fat_percentage	Numeric	%	Apple (Body Fat Percentage), Google (Body Fat), Samsung/Garmin (Body Composition)	Estimated percentage of adipose tissue.
weight_and_bmi	Numeric	kg/index	Apple (BMI, Lean Body Mass), Google (Weight, Height)	Body mass and structural measurements.
nutrition_log	String/JSON	N/A	Samsung (Nutrition), Google (Nutrition Log)	Input detailing dietary and hydration entries.

References

- <https://developer.apple.com/documentation/healthkit/data-types>
- <https://developer.samsung.com/health/data/guide/features/data-types.html>
- <https://developers.google.com/health/data-types>
- <https://developer.garmin.com/health-sdk/overview/>